Type MS · Identify a measurement weakness and improve accuracy

Answer 1 of 6 · Question 1 part a.iv · 3 marks

2021 NOV · Paper 01 · extract 1 of 2 · find part a.iv in this mark scheme extract

Question number				
1	(a)	(i)	<table><tr><th>Conditions</th><th>Result/no/scale of bubbles</th></tr><tr><td>Room temperature</td><td>5</td></tr><tr><td>Alkali added</td><td>3</td></tr><tr><td>Acid added</td><td>2</td></tr><tr><td>3°C</td><td>1</td></tr><tr><td>70°C</td><td>0</td></tr><tr><td>no hydrogen peroxide added</td><td>0</td></tr></table> • two column table; • headings; • correct order of results (high to low/ORA);	3 marks
		(ii)	• no repeats/only done once;	1 mark
		(iii)	• results based on judgement/subjective/qualitative;	1 mark
		(iv)	• repeats; • measure volume of oxygen; • using gas syringe/other suitable apparatus;	3 marks
	(b)	(i)	untreated/room temperature;	1 mark
		(ii)	control;	1 mark
		(iii)	• reduced/low enzyme activity; • not at optimum pH; • active site changed; • fewer enzyme substrate complexes/substrate won't fit;	4 marks

Total 14 marks

Type MS · Identify a measurement weakness and improve accuracy

Answer 1 of 6 · Question 1 part a.iv · 3 marks

2021 NOV · Paper 01 · extract 2 of 2 · find part a.iv in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Identify a measurement weakness and improve accuracy

Answer 2 of 6 · Question 1 part a.iii · 1 mark

2021 NOV · Paper 01 · extract 1 of 2 · find part a.iii in this mark scheme extract

Number				
1	(a) (i)	<table><tr><th>Conditions</th><th>Result/no/scale of bubbles</th></tr><tr><td>Room temperature</td><td>5</td></tr><tr><td>Alkali added</td><td>3</td></tr><tr><td>Acid added</td><td>2</td></tr><tr><td>3°C</td><td>1</td></tr><tr><td>70°C</td><td>0</td></tr><tr><td>no hydrogen peroxide added</td><td>0</td></tr></table> • two column table; • headings; • correct order of results (high to low/ORA);		3 marks
	(ii)	• no repeats/only done once;		1 mark
	(iii)	• results based on judgement/subjective/qualitative;		1 mark
	(iv)	• repeats; • measure volume of oxygen; • using gas syringe/other suitable apparatus;		3 marks
	(b) (i)	untreated/room temperature;		1 mark
	(ii)	control;		1 mark
	(iii)	• reduced/low enzyme activity; • not at optimum pH; • active site changed; • fewer enzyme substrate complexes/substrate won't fit;		4 marks

Total 14 marks

Type MS · Identify a measurement weakness and improve accuracy

Answer 2 of 6 · Question 1 part a.iii · 1 mark

2021 NOV · Paper 01 · extract 2 of 2 · find part a.iii in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Identify a measurement weakness and improve accuracy

Answer 3 of 6 · Question 7 part b.iv · 1 mark

2020 NOV · Paper 02 (regional R) · extract 1 of 2 · find part b.iv in this mark scheme extract

Number					
7	(a)	(i)	• allows protein to stay longer in stomach/gut (1) • gives time for full digestion (1)		2
		(ii)	• diet largely/entirely milk when young (1) • changes to solids when older (1)		2
		(iii)	• biuret test (1) • add reagent to solid (1) • blue to lilac/pale purple (1)		3
	(b)	(i)	• temperature (1) • volume of milk (1) • volume/concentration of chymosin (1)		3
		(ii)	$\frac{238 + 232 + 241 + 229}{4}$ (1) = 235 (1)		2
		(iii)	not included when calculating mean (1)		1
		(iv)	bubbles of different sizes/volume of CO_2 variable (1)		1

Total for this question = 14 marks

Type MS · Identify a measurement weakness and improve accuracy

Answer 3 of 6 · Question 7 part b.iv · 1 mark

2020 NOV · Paper 02 (regional R) · extract 2 of 2 · find part b.iv in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Identify a measurement weakness and improve accuracy

Answer 4 of 6 · Question 7 part b.ii · 2 marks

2019 JUN · Paper 01 (regional R) · extract 1 of 3 · find part b.ii in this mark scheme extract

7(a)	Any four from: • volume of lactose/sugar solution used; • concentration of lactose/sugar solution; • volume of enzyme solution used; • concentration of enzyme; • type of enzyme used; • how to measure amount/concentration of lactic acid / apparatus to measure amount / concentration of lactic acid;	Allow amount	**(4)**

Question Number	Answer	Additional guidance	Mark
7(b)(i)	25 °C;	accept 20 °C to 45 °C	**(1)**

Question Number	Answer	Additional guidance	Mark
7(b)(ii)	Any two from: • too few results/temperatures investigated / temperatures are too far apart/large gaps between temperatures; • peak/maximum lactic acid production could be below 25°C / could be above 25°C;		**(2)**

Question Number	Answer	Additional guidance	Mark
7(c)	Any three from: • little/no increase in lactic acid concentration / lactic acid concentration would reach a maximum/peak/become constant; • enzyme denatures (due to decrease in pH); • all substrate converted/broken down/used up; • so reaction /slows down/stops;		**(3)**

Type MS · Identify a measurement weakness and improve accuracy

Answer 4 of 6 · Question 7 part b.ii · 2 marks

2019 JUN · Paper 01 (regional R) · extract 2 of 3 · find part b.ii in this mark scheme extract

Question Number	Answer	Additional guidance	Mark
7(d)	Any five from: • not possible (to be completely certain); • due to (low) number of pH/temperature values tested/more pH/temperature values need to be tested; • pH5 appears to be optimum pH; • because most sugar broken down/converted at this pH; • optimum pH could be between 5-8/2-5; • optimum temperature not known; • no repeats carried out/no means calculated; • anomalies not identified; • there could be other components in milk that increase enzyme activity/control variables not taken into account;	Allow value stated within ranges given	**(5)**

Total 15 Marks

Type MS · Identify a measurement weakness and improve accuracy

Answer 4 of 6 · Question 7 part b.ii · 2 marks

2019 JUN · Paper 01 (regional R) · extract 3 of 3 · find part b.ii in this mark scheme extract

Paper 1: Subject Human Biology

Question Number	Answer	Additional guidance	Mark

Type MS · Identify a measurement weakness and improve accuracy

Answer 5 of 6 · Question 6 part a.iii · 2 marks

2023 JAN · Paper 01 · extract 1 of 2 · find part a.iii in this mark scheme extract

Number					
6	(a)	(i)	18-12; 6;	accept 17.8/17.9	2
		(ii)	Two from: • volume of milk; • volume of the starter culture/live yoghurt; • type of starter culture • type of milk; • type of live yoghurt; • mixing;	reject 'amount'	2
		(iii)	• repeat the investigation at more temperatures; • between 20°C and 60°C;		2
		(iv)	• heated to kill (unwanted) bacteria in the milk; • cooled so that the bacteria in the starter culture are not killed;		2
	(b)		Any two from: • pH 4.4 inhibits the bacteria/bacterial growth; • denatures enzymes; • changes the shape of the active sites • lactose cannot bind to the enzyme/no enzyme-substrate complexes formed;		2
	(c)	(i)	Any two from: • the beads can be re-used; • product doesn't need to be treated to remove the enzyme; • large surface area to increase the rate of reaction; • enzyme more temperature stable;		2
		(ii)	• long/villi/microvilli to increase surface area; • concentration gradient maintained with blood flow; • short diffusion distance;		3

Total 15 marks

Type MS · Identify a measurement weakness and improve accuracy

Answer 5 of 6 · Question 6 part a.iii · 2 marks

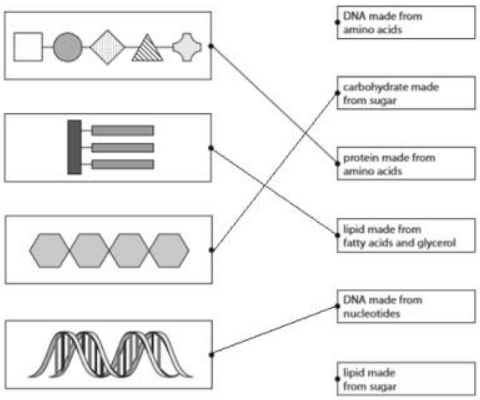
2023 JAN · Paper 01 · extract 2 of 2 · find part a.iii in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Identify a measurement weakness and improve accuracy

Answer 6 of 6 · Question 3 part c.ii · 1 mark

2019 JUN · Paper 02 · extract 1 of 2 · find part c.ii in this mark scheme extract

3	(a)	( ) 	Do not allow more than one line from each structure	3
	(b)	**D** stomach;		1
	(c) (i)	3/three		1
	(ii)	not washing equipment/transfer of food containing protein into distilled water;	Allow any valid alternative	1
	(d)	Hazard: Broken glass - cuts Reducing risk: Clear up breakages using a dustpan and brush/keep equipment away from edge of tables;	Ignore wear gloves	1
		Hazard: Biuret reagent - irritant Reducing risk: Wear goggles/wash hands after use/don't use if skin is sensitive;	Ignore wear gloves	1

Total question 3 = 8 marks

Type MS · Identify a measurement weakness and improve accuracy

Answer 6 of 6 · Question 3 part c.ii · 1 mark

2019 JUN · Paper 02 · extract 2 of 2 · find part c.ii in this mark scheme extract

Question number	Answer	Notes	Marks